Security Attack Detection Using Artificial Neural Network and Random Forest

Samaresh — PES2UG24CS657
B S SHARATH CHANDRA REDDY — PES2UG24CS630
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1. Problem Statement

Network intrusion detection systems must identify malicious network behaviour from connection-level information. This project reproduces the Artificial Neural Network (ANN) approach presented in the reference paper "Using Neural Network for Security Attacks Detection" using the NSL-KDD dataset.

The first objective is to reproduce the preprocessing, ANN architecture, and training configuration described in the reference paper and compare the obtained results with the reported results. The second objective is to extend the approach by implementing a Random Forest classifier and comparing its performance with the reference ANN.

2. Dataset and Preprocessing

The project uses the NSL-KDD dataset, which contains 41 features for each network connection record. The attacks are organized into four major categories: DoS, Probe, R2L, and U2R, along with normal network behaviour. The reference paper also notes that the test dataset contains attack types that are not present in the training dataset.

The preprocessing pipeline follows the reference paper. The categorical features protocol_type, service, and flag were one-hot encoded. The resulting feature representation contains 122 input features, matching the dimensionality reported in the paper. Numerical features were standardized using mean-zero and unit-variance scaling.

The target was represented using 24 output classes: 23 classes corresponding to the normal record and known attack types, plus one unknown class for attack types that are not present in the training data. This allows previously unseen test attacks to be represented during evaluation.

3. Reference ANN Implementation

The reproduced ANN follows the architecture described in the reference paper:

122 input units → 500 hidden units → 24 output units

The hidden layer uses the sigmoid activation function and the output layer uses softmax. The model was trained using a learning rate of 0.5, batch size of 1000, and 10 epochs.

For reproducibility, a fixed random seed was used in the implementation. This is particularly relevant because the reference paper states that repeated training runs can produce different results due to random initialization of the neural-network parameters.

Note: The reference paper specifies a regularized mini-batch gradient ascent algorithm with a learning rate of 0.5, batch size of 1000, and 10 epochs. However, the paper does not specify the regularization coefficient or the exact regularization implementation. Therefore, our implementation uses standard SGD with the reported learning rate, batch size, and number of epochs, without introducing an unspecified regularization term.

4. ANN Results

The reproduced ANN achieved:

Metric	ANN
Test Accuracy	68.77%
Normal/Abnormal Accuracy	75.47%
Macro F1	0.36
Weighted F1	0.59
Known-Class Accuracy	82.49%

The reference paper reports approximately 70% test accuracy and 80% Normal/Abnormal accuracy, compared with 68.77% and 75.47% respectively in our implementation. The reproduced results are reasonably close to the reported performance.

5. Additional Model — Random Forest

Random Forest was selected as the additional model because it provides a fundamentally different classification approach and is suitable for structured tabular data.

The Random Forest classifier was configured with 200 decision trees and random_state = 42. It was trained using the same preprocessed dataset and evaluated using the same test dataset and evaluation metrics as the ANN.

The Random Forest achieved:

Metric	Random Forest
Test Accuracy	72.21%
Normal/Abnormal Accuracy	74.57%
Macro F1	0.44
Weighted F1	0.62
Known-Class Accuracy	86.61%

6. Model Comparison and Conclusion

Random Forest achieved better performance than the reference ANN for detailed multi-class attack classification. Its test accuracy increased from 68.77% to 72.21%, while Macro F1 increased from 0.36 to 0.44, Weighted F1 from 0.59 to 0.62, and known-class accuracy from 82.49% to 86.61%.

For the Normal/Abnormal classification task, the ANN achieved slightly higher accuracy (75.47%) than Random Forest (74.57%).

Overall, the reference ANN was successfully reproduced using the NSL-KDD dataset and its reported architecture. The additional Random Forest model provided stronger performance for multi-class attack classification, demonstrating the usefulness of comparing different machine-learning approaches for intrusion detection.